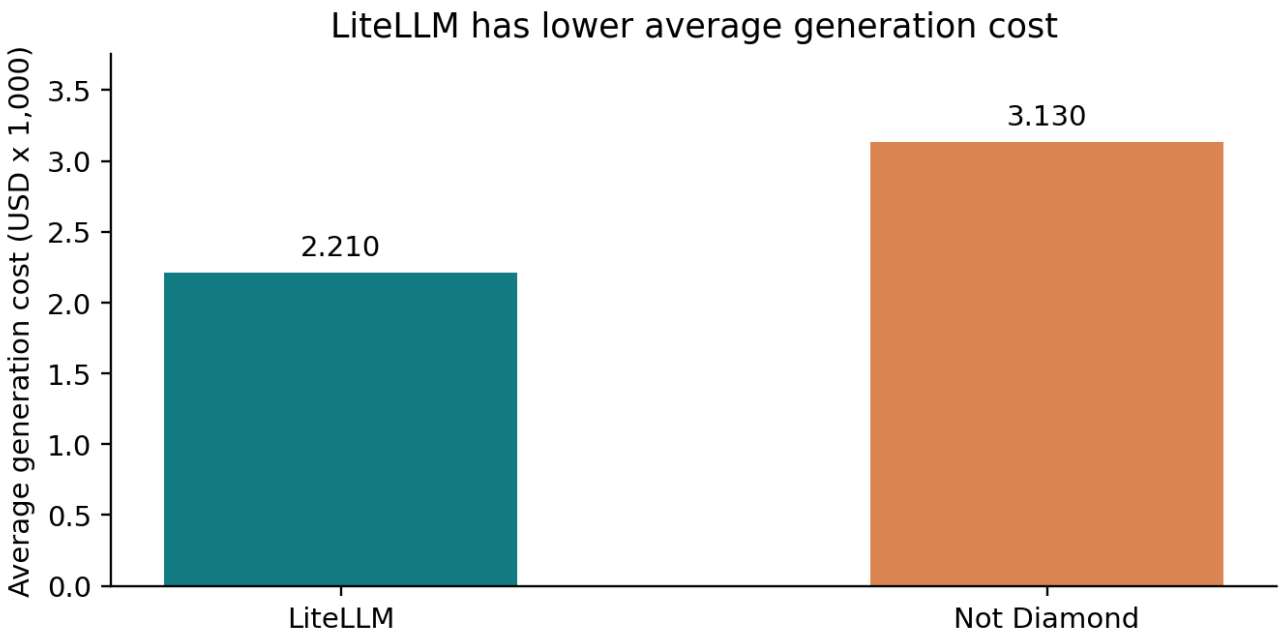
Stage A | Router Selection Report

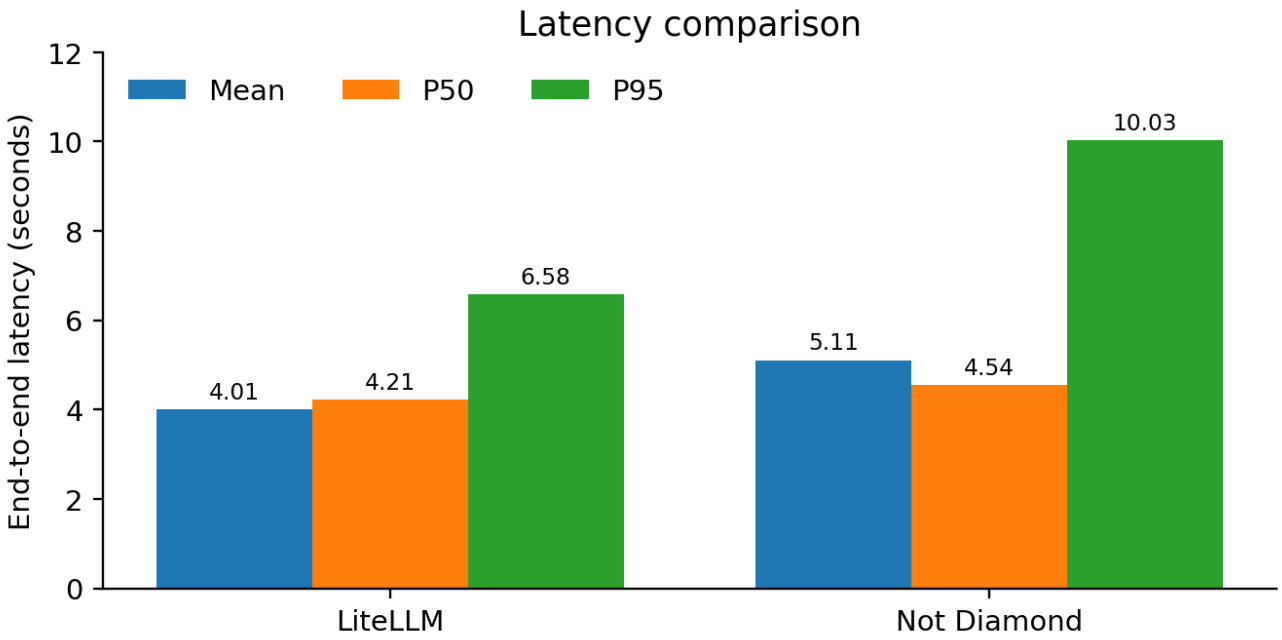
Held-out evaluation: LiteLLM vs Not Diamond | 7 October 2026

Decision: Select **LiteLLM** as the Stage B baseline. Across 20 matched held-out tasks, it achieved 29.4% lower mean generation cost and 21.4% lower mean end-to-end latency, with both routers passing all judge-scored outputs.

Metric	LiteLLM	Not Diamond
Successful generations	20 / 20	20 / 20
Mean judge score	1.0000	0.9958
Judge pass rate	100%	100%
Mean generation cost	$0.002210	$0.003130
Total generation cost	$0.044196	$0.062607
Mean latency	4.01 s	5.11 s
Median latency	4.21 s	4.54 s
P95 latency	6.58 s	10.03 s

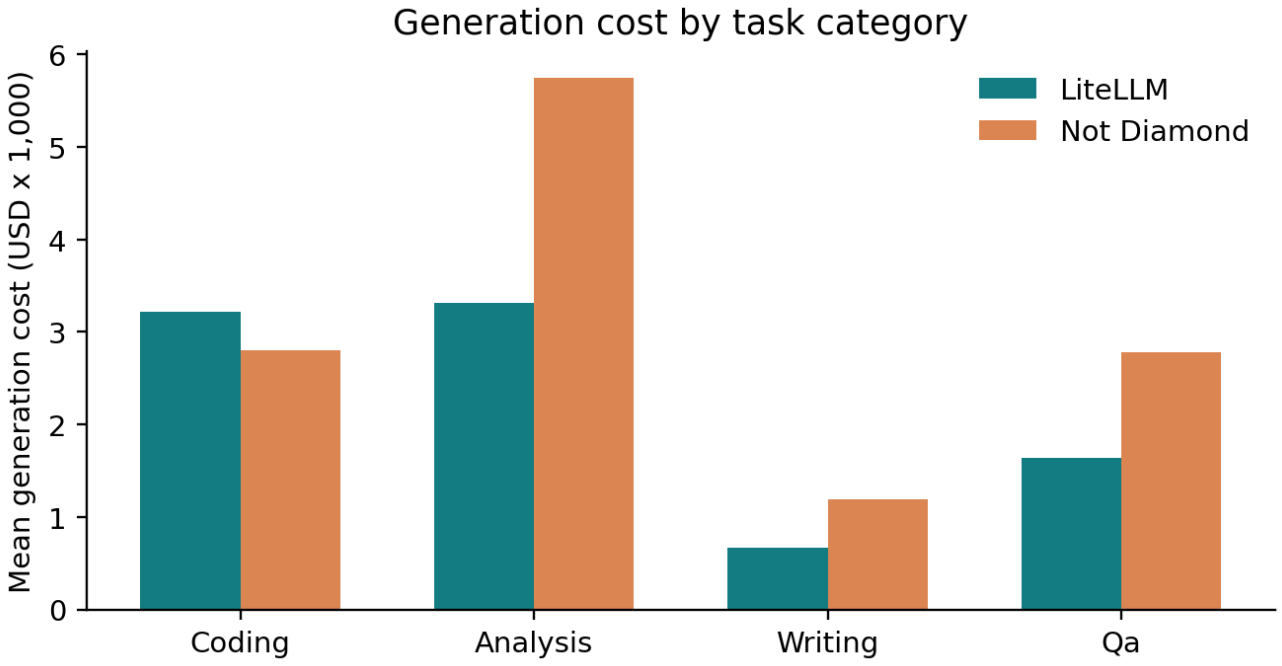
1. Cost and latency

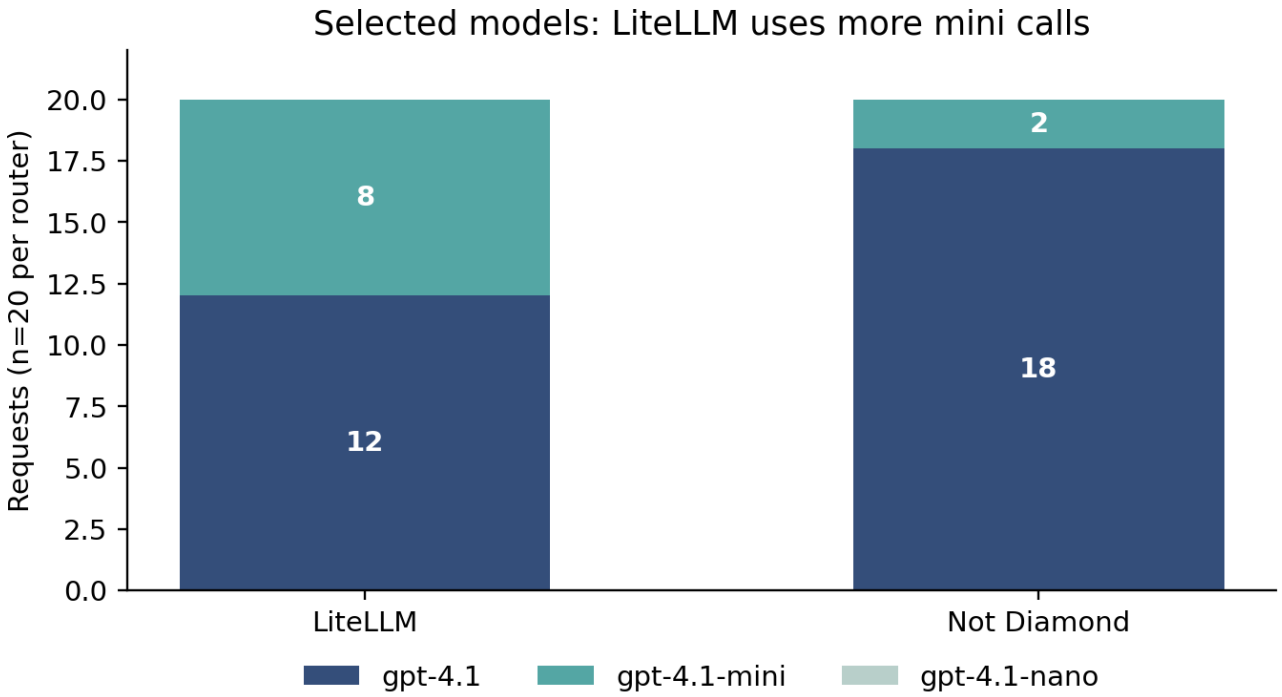




Costs are generation-only; router service fees and judge costs are excluded. Latency measures the complete routing and generation request.

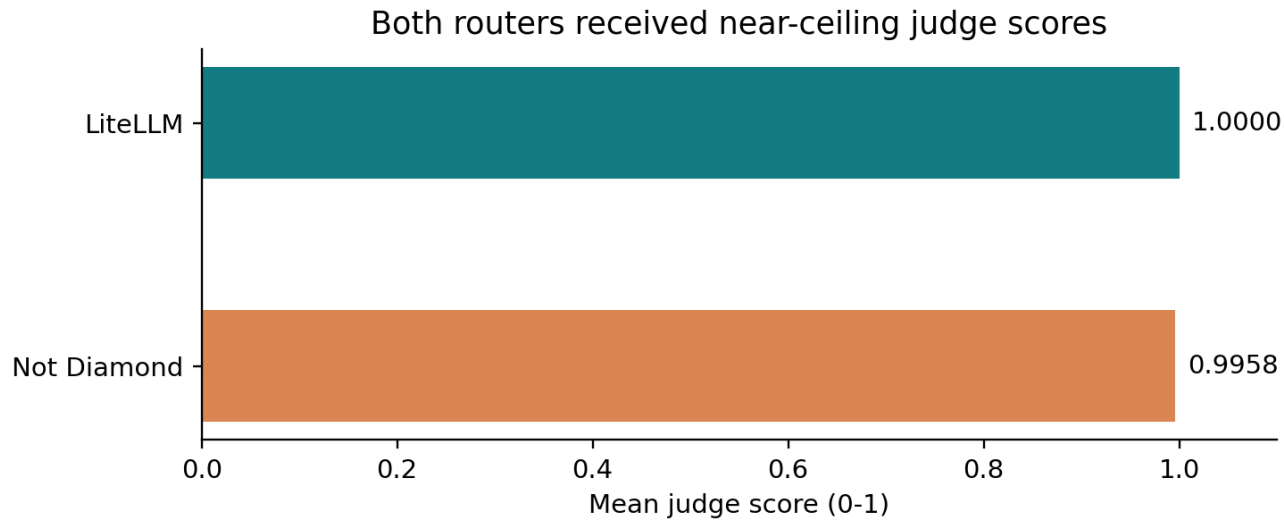
2. Task-level and model-selection patterns

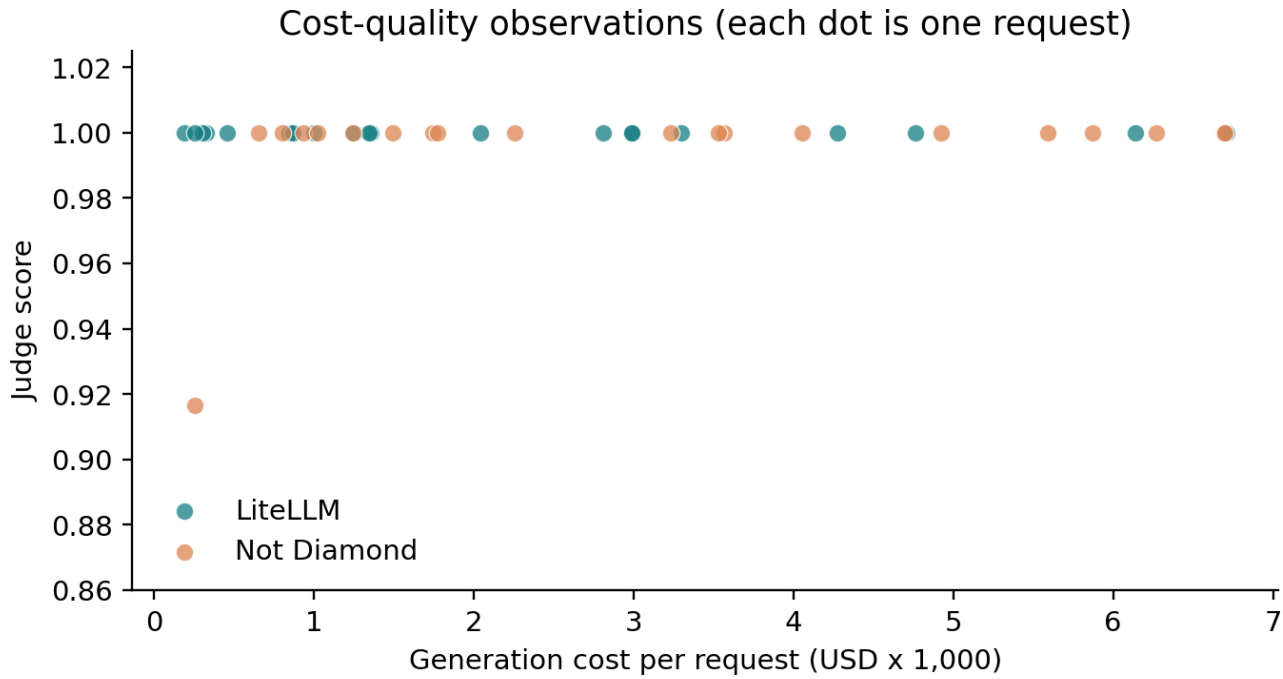




The candidate pool was identical for both routers: GPT-4.1, GPT-4.1 mini, and GPT-4.1 nano. The distributions above reflect the actual selected models in the held-out run.

3. Quality and cost-quality tradeoff





Quality was scored by GPT-4.1 mini, blinded to router identity and selected model, using correctness, completeness, and instruction-following (1-5 each). Scores were normalized to 0-1; passing threshold = 0.75. One Not Diamond response scored 0.9167; all other responses scored 1.0.

4. Interpretation and limitations

Why LiteLLM: It delivered lower average cost and latency while preserving the same observed judge pass rate. The cost difference partly reflects different underlying model selections, not only router overhead.

Limitations: This is one small held-out run (20 paired tasks), not a statistical equivalence or significance study. The LLM judge exhibits a strong ceiling effect and may favor same-family outputs. LiteLLM is adaptive; router state and run order may affect selections. Generation prices exclude routing-service charges. Coding answers were not independently executed against deterministic tests.